

URBAN POLICY DELIVERY CHAIN

Bristol One City V1.1

7-phase N-D-C analysis of Bristol's One City Plan delivery chain: from political mandate through governance, capital deployment, and infrastructure delivery to measurable city-wide outcomes. Four missions. 2030 carbon-neutral target. Municipal sustainability-linked bond opportunity.

81.3%

Chain balance (Good zone; masked by Phase 5 bottleneck)

58.3%

Phase 5 balance: community uptake (Poor: primary bottleneck)

10.7 yr

Male life expectancy gap: most vs. least deprived wards

£1bn+

City Leap 20-yr JV investment vs. £2m BCAI raised to date

PRIMARY STRUCTURAL FAILURE: COMMUNITY AND HOUSEHOLD UPTAKE (PHASE 5, 58.3%)

Bristol has built an exceptionally well-defined delivery system at the institutional level: the One City Plan has strong mandates (Phase 0, 88.9%), coordinated governance (Phase 1, 84.0%), contractually-grounded strategies (Phase 2, 90.6%), and robust outcome measurement (Phase 6, 91.9%). The failure is at the point of contact with communities. Only 750 of 470,000 residents participate in the Bristol Climate Action Investment. The 10.7-year male life expectancy gap between deprived and affluent wards is not improving. Scope 3 (consumption-based) emissions account for 60–75% of a typical Bristol household's carbon footprint and 70–90% of the city's corporate footprint, yet remain structurally untracked in the delivery chain. The D-C gap at Phase 5 is 100 points: the widest in the chain. Capital deployment (Phase 3, 70.9%) is the secondary bottleneck: £1bn+ City Leap institutional framework vs. £2m mobilised through citizen mechanisms against a city economy of £22.8bn GDP.

MUNICIPAL FINANCE CONTEXT

Source: Bristol City Council Budget Report 2026/27; Wikipedia Economy of Bristol; BBC News Jun 2025. Added v1.1.

£22.8bn

City GDP (£47,148 per capita; pop. 482,998). Strongest performer among major UK cities outside London.
Wikipedia / Centre for Cities 2024

£63m

SEND / Dedicated Schools Grant deficit "time bomb": council faces insolvency risk within 10 months without central government extension.
BBC News Jun 2025

£634.1m

Bristol City Council General Fund budget 2026/27, funded via council tax (Band D: ~£2,460), business rates, and government grants.
BCC Budget Report 2026/27

£32m/yr

Annual Private Finance Initiative (PFI) charges servicing long-term infrastructure assets (Temple Quarter regeneration and others).
BCC Capital Strategy 2023/24–2032/33

CORPORATE SCOPE 3 (GHG PROTOCOL)

Represents **70–90%** of an average company's total GHG inventory. In Commercial Real Estate and Construction, indirect embodied carbon and tenant energy push this to **85–95%**. Framework: GHG Protocol Corporate Value Chain Standard (15 categories). Estimation: spend-based modelling, activity metrics, or LCA.

RESIDENTIAL SCOPE 3 (HOUSEHOLD)

Approximately **60–75%** of a Bristol household's personal carbon footprint comes from indirect (Scope 3) sources: embedded emissions of food, clothing, purchased goods, and services. Direct gas heating = Scope 1; electricity = Scope 2. Framework: PAS 2070 or territorial consumption-based accounting. Estimation: household expenditure surveys, localized utility and consumption data.

ANALYTICAL LAYERS



Supply Chain

7-phase urban delivery chain from political mandate through governance, strategy, capital, infrastructure, community uptake to outcome measurement. Phase 5 bottleneck analysis. N-D-C dashboard, project context, and intervention worksheet.



Value Chain

Municipal SL Bond structure under evaluation: cTVF KPI calibration against Bristol City Council financials (£634.1m budget, £63m SEND deficit) and City Leap carbon/energy targets. ICMA SLBP alignment, SPT design, and coupon step-up structure. City Leap Business Plan (Apr 2024) provides energy and carbon KPI anchors. Requires Phase 5 intervention data and institutional investor engagement before activation.

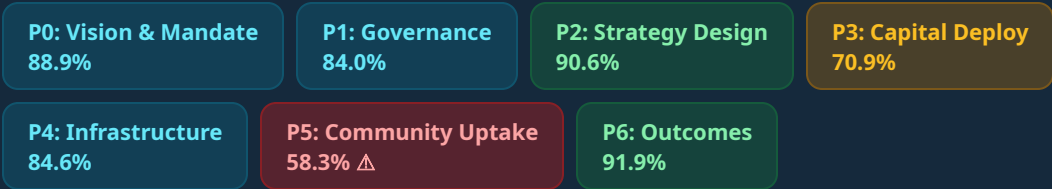


Just Transition Return Chain

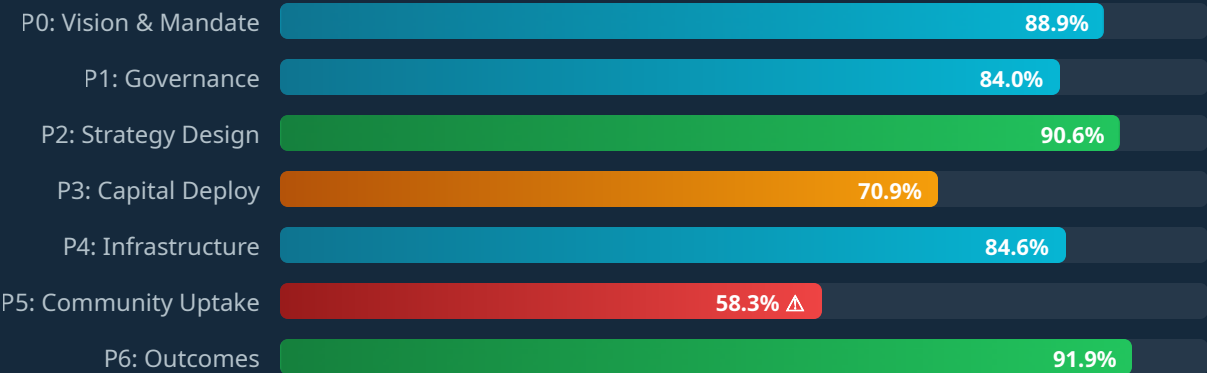
TVPCI-R analysis: how much of City Leap, BCAI, and One City investment recirculates to Bristol's most deprived communities. B_chain = 47.0 (severe imbalance). TVPCI-R = 50.7 (Failure zone).

PHASE HEALTH MAP

7 delivery phases: Phase 5 is the primary bottleneck

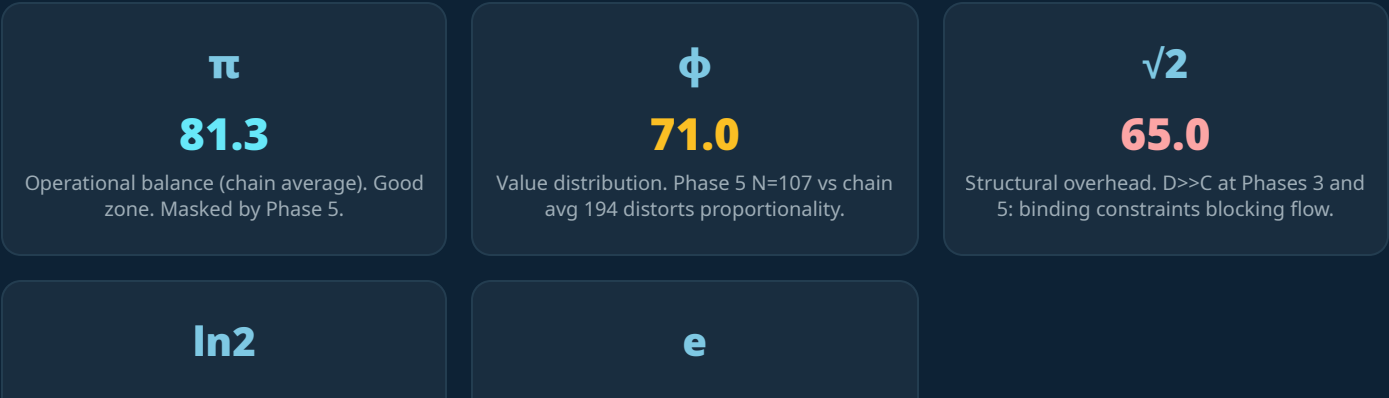


PHASE N-D-C BALANCE (SYNTHETIC BASELINE)



Thresholds: Excellent ≥95% | Good 80-94% | Fair 61.8-79% | Poor <61.8%

FIVE-MODEL COHERENCE (BASELINE)



61.0

Temporal gap, 6+ years from 2019 declaration; 2030 deadline is 4 years away.

52.0

Financial abstraction. £2m BCAI vs £22.8bn city GDP. Municipal SL Bond structure under cTVF evaluation (v1.1).

KEY STAKEHOLDERS

GOVERNANCE BOARD

City Funds (Andy Street)
Bristol City Council (Christina Gray, Nick Hibbert, Tony Dyer)
TUC (Ines Lage)
University of Bristol (2 seats)
City of Bristol College
Bristol Climate and Nature Partnership (Lizzie Testani)
University Hospitals Bristol and Weston NHS
North Bristol NHS Trust
VOSCUR (voluntary sector)

DELIVERY PARTNERS

Ameresco Ltd (City Leap 50% JV partner)
Vattenfall Heat UK (City Leap subcontractor)
Abundance Investment (BCAI platform)
Arup (Climate Strategy author)
Centre for Sustainable Energy
Cabot Institute, University of Bristol (VLR)
Green Finance Institute (LCB toolkit)
Living Places Earth (place-based advisor)
Bristol Advisory Committee on Climate Change

FINANCIAL AND BOND LAYER

Bristol City Council (50% City Leap stake)
PWLB (primary borrowing facility)
Abundance Investment (LCB issuance)
Institutional bond market (SLB potential)
City Funds (impact capital pool)
DESNZ (Department for Energy Security and Net Zero)
ICMA (SLB Principles framework)
TrueValue Analytics (cTVF verification)

← TrueValue Analytics Home
Just Transition Return Chain

Supply Chain Analysis
Value Chain (Deferred)
Budget Report 2026/27

N-D-C Dashboard
City Leap Business Plan (PDF)

Project Context
BCC

SUPPLY CHAIN ANALYSIS

Bristol One City: Urban Delivery Chain

7-phase delivery chain analysis. Chain balance 81.3% (Good). Primary bottleneck: Phase 5 Community and Household Uptake (58.3%, Poor). The city has exemplary institutional infrastructure but a critical disconnect at the point of contact with deprived communities.

81.3%

Chain average balance (Good)

194

Average N score (chain coherence)

2

Bottleneck phases (P5, P3)

100 pts

D-C gap at Phase 5 (largest in chain)

ANALYSIS PAGES



N-D-C Dashboard

Phase-level D, C, N bars; coherence pentagon; bottleneck alert; full phase data table.



Project Context

Geographic and governance scope; bond opportunity analysis; Phase Intervention Worksheet; data sources.



Just Transition Return Chain

TVPCI-R community wealth recirculation analysis. TVPCI-R = 50.7 (Failure). B_chain = 47.0.

PHASE MAP: 7-PHASE URBAN DELIVERY CHAIN

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	KEY CONSTRAINT OR OUTPUT
P0	Vision and Mandate	270	240	88.9%	226	GOOD	HIGH	5 plan iterations; 2030 net-zero; 4 missions; 17 SDGs; BACCC
P1	Governance and Coordination	250	210	84.0%	192	GOOD	MEDIUM	7 thematic boards; Governance

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	KEY CONSTRAINT OR OUTPUT
								Board (12 members); BACCC independent review
P2	Strategy and Program Design	265	240	90.6%	228	EXCELLENT	HIGH	City Leap binding KPIs; IBP £631m; BCAI Green Finance Framework; LNRS
P3	Capital Deployment	275	195	70.9%	164	FAIR	MEDIUM	£1.67bn capital budget; BCAI only £2m raised; scope 3 capital near-zero
P4	Infrastructure Delivery	260	220	84.6%	202	GOOD	HIGH	City Leap KPIs on track; heat network; solar at schools; housing retrofit; 10% BNG
P5 	Community and Household Uptake	240	140	58.3%	107	POOR	LOW	750 BCAI investors from 470k population; 10.7yr life expectancy gap; scope 3 untracked
P6	Outcome Measurement	270	248	91.9%	238	EXCELLENT	HIGH	256 QoL indicators; PHOF; City Leap KPI audit; VLR; quarterly

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	KEY CONSTRAINT OR OUTPUT
								Power BI dashboard

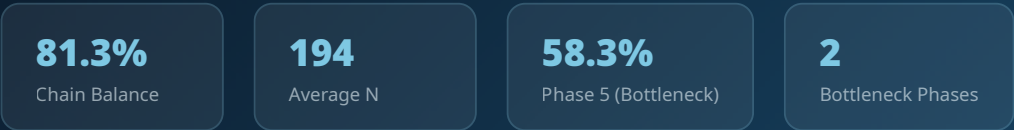
N-D-C FRAMEWORK APPLIED TO URBAN POLICY DELIVERY

In the Tholonic N-D-C framework, **D (Definition)** captures the constraints, requirements, governance obligations, and binding commitments that define what each delivery phase must achieve. **C (Contribution)** captures the actual outputs, measurable delivery, and flows produced by each phase. **N (Negotiation)** is the coherent emergent state of the phase: the product of D and C in balance.

For Bristol, D across Phases 0-2 is very high: the city has invested heavily in policy commitments, governance structures, and binding strategies. The D-C gap widens at Phase 3 (capital) and collapses into a structural failure at Phase 5 (community uptake). Phase 6 (measurement) recovers because Bristol's monitoring infrastructure is genuinely world-class: the QoL survey (256 indicators), BACCC independent scrutiny, and City Leap contractual audit form an unusually robust verification layer. This measurement strength is the foundation on which a Sustainability-Linked Bond KPI framework can be built.

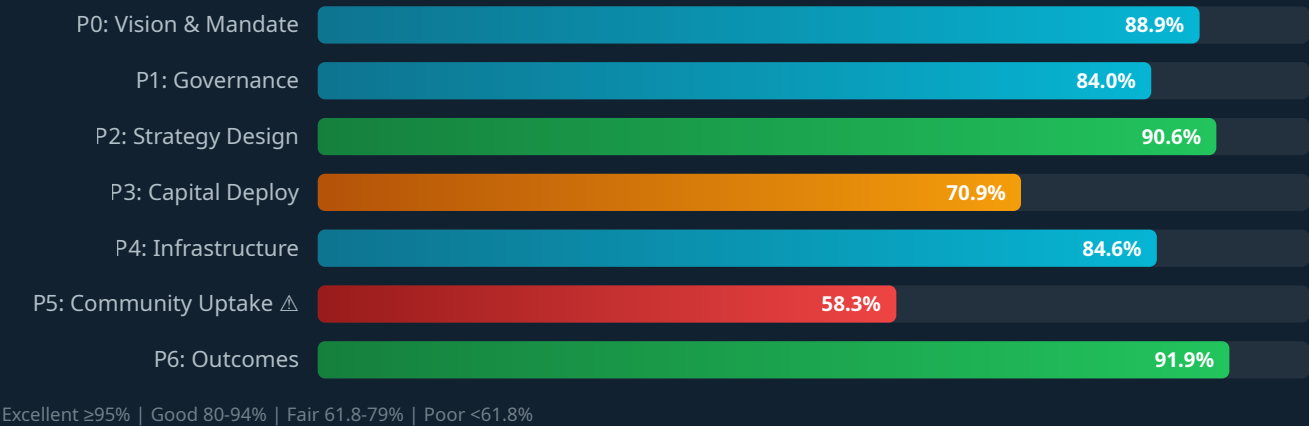
Bristol One City: N-D-C Dashboard

Urban Policy Delivery Chain: 7 phases: Synthetic baseline, PROVISIONAL

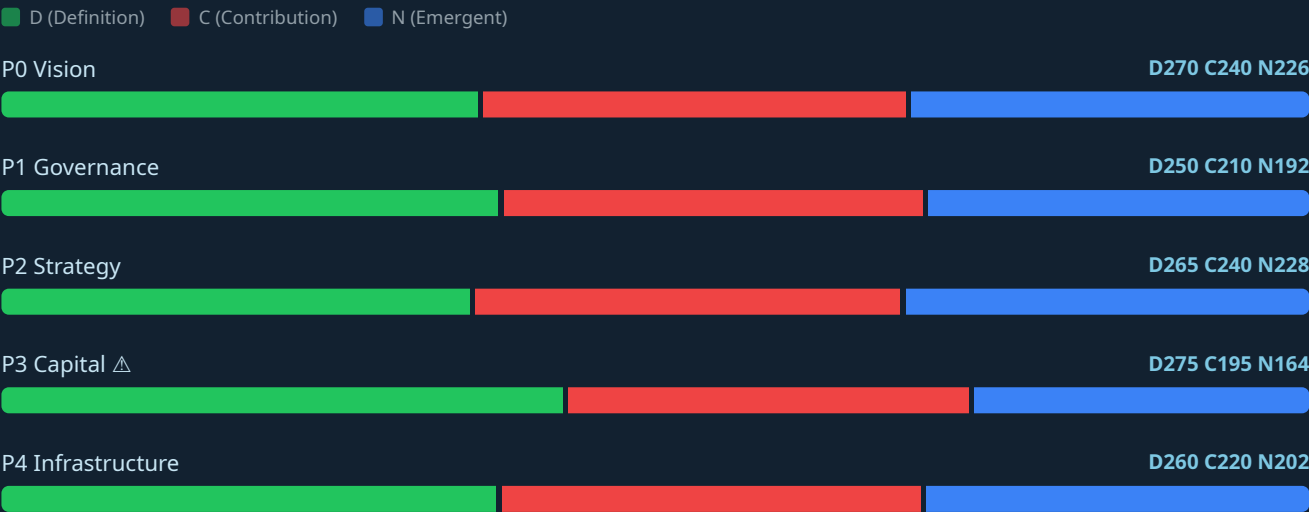


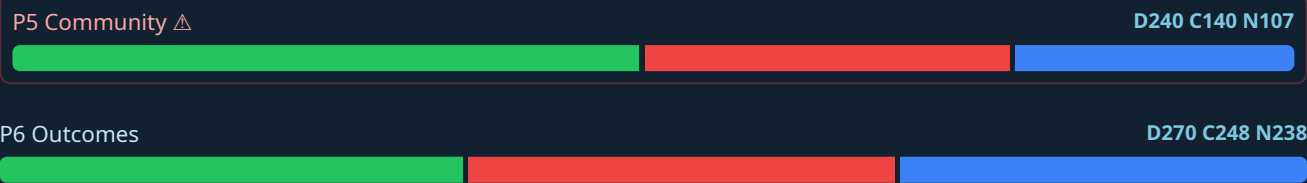
Bottleneck: Phase 5: Community and Household Uptake (58.3%, Poor). D-C gap of 100 points: 240 (requirements, designed programmes) vs 140 (actual community delivery). Only 750 of 470,000 residents are BCAI investors. The 10.7-year male life expectancy gap between most and least deprived wards is not improving. Scope 3 emissions are structurally untracked. Secondary bottleneck: Phase 3 Capital Deployment (70.9%): £1.67bn capital budget vs £2m mobilised through citizen investment mechanisms.

PHASE BALANCE SCORES



D / C / N PER PHASE (NORMALISED TO 300)





FIVE-MODEL COHERENCE (BASELINE)

π
81.3

Operational balance: chain average. Good zone but Phase 5 is masked.

ϕ
71.0

Value distribution: Phase 5 N=107 vs chain avg 194 distorts proportionality.

$\sqrt{2}$
65.0

Structural overhead: D>>C at P3 (gap 80pts) and P5 (gap 100pts).

$\ln 2$
61.0

Temporal gap: 6+ years since 2019 declaration; 2030 deadline 4 years away.

e
52.0

Financial abstraction: £2m BCAI vs £1bn+ needed. No SLB structured yet.

FULL PHASE DATA TABLE

PH	NAME	D	C	D-C GAP	BALANCE	N	STATUS	TRANSPARENCY	DATA QUALITY	KEY METRIC
P0	Vision and Mandate	270	240	30	88.9%	226	GOOD	HIGH	HIGH	5 plan iterations; 2030 net-zero (scope 1/2/3); 200+ stakeholders consulted
P1	Governance	250	210	40	84.0%	192	GOOD	MEDIUM	MEDIUM	7 boards; 12-member Governance Board; BACCC 12 recommendations (2023)
P2	Strategy Design	265	240	25	90.6%	228	EXCELLENT	HIGH	HIGH	City Leap: 140,000t carbon, 180MW, 327GWh KPIs; IBP £631m; LNRS (Nov 2024)
P3	Capital Deployment	275	195	80	70.9%	164	FAIR	MEDIUM	MEDIUM	£1.67bn capital budget; BCAI £2m raised; £30m of £61.5m social value target delivered

PH	NAME	D	C	D-C GAP	BALANCE	N	STATUS	TRANSPARENCY	DATA QUALITY	KEY METRIC
P4	Infrastructure	260	220	40	84.6%	202	GOOD	HIGH	HIGH	City Leap KPIs on track; council direct emissions -67% since 2015; 10% BNG mandatory
P5 ⚠	Community Uptake	240	140	100	58.3%	107	POOR	LOW	LOW	750 BCAI investors / 470k pop; life expectancy gap 10.7yr (men); scope 3 untracked
P6	Outcomes	270	248	22	91.9%	238	EXCELLENT	HIGH	HIGH	256 QoL indicators; PHOF; City Leap KPI audit; VLR; quarterly Power BI (public)
All values PROVISIONAL. balance=100-(D-C /max(D,C))*100; N=sqrt(D*C)*(balance/100)										

SUPPLY CHAIN: PROJECT CONTEXT

Bristol One City: Urban Delivery Chain

7-phase N-D-C analysis of Bristol's One City Plan delivery chain. 470,000 population. 2030 carbon-neutral target (scope 1/2/3). Four missions. City Leap £1bn+ 20-year joint venture. Bristol Climate Action Investment (4.2% LCB). Municipal sustainability-linked bond opportunity. TVPCI = 81.3% (Good zone). Bottleneck: Phase 5 (58.3%, Poor).

81.3%

Chain balance (Good; Phase 5 masked)

58.3%

Phase 5 balance (primary bottleneck)

10.7 yr

Male life expectancy gap (most vs least deprived)

52.0

e-axis score (financial abstraction: SLB gap)

City Scope and Context

Bristol is a city of approximately 470,000 people in South West England, governed by Bristol City Council and coordinated across sectors by the Bristol One City platform. The city declared a climate emergency in 2018, was the first UK local council to do so, and set a net-zero target for 2030 covering scope 1, 2, and 3 emissions. Bristol One City is not a council body: it is a cross-sector platform convened by the City Office, coordinating seven thematic boards and a Governance Board drawn from anchor institutions across public, private, voluntary, and academic sectors.

PARAMETER	VALUE	SOURCE
Population	~470,000	Bristol Key Facts 2025
Carbon neutral target	2030 (scope 1, 2, and 3)	One City Climate Strategy (Arup, 2020)
CO2 reduction since 2005	43% (city-wide)	Bristol City Council Q3 2023-24 Performance
Council direct emissions reduction	67% since 2015	Abundance Investment / BBC News (March 2025)
Most deprived population (IMD 2025)	53,100 people (11%) in 10% most deprived nationally	Bristol Key Facts November 2025
Male life expectancy gap	10.7 years (most vs least deprived, 2021-23)	PHOF 2025 (A02a)
City Leap total investment (20yr)	>£1 billion	Bristol City Leap website

PARAMETER	VALUE	SOURCE
City Leap 5yr IBP investment	£631 million	City Leap Initial Business Plan (April 2024)
BCAI raised (citizen investment)	£2 million (2 tranches, 750+ investors)	Abundance Investment / Room151 (2025)
Capital budget 2026-27	£1,669.3 million (approved)	Bristol City Council Budget 2026-27
One City Plan iterations	5 (2019, 2020, 2021, 2023, 2025)	Bristol One City website

Phase Map: 7-Phase Urban Delivery Chain

The Bristol One City delivery chain is modelled as a 7-phase system from political mandate through community uptake to measured outcomes. Unlike a physical commodity chain, D (Definition) represents the governance requirements, policy commitments, and binding constraints at each phase. C (Contribution) represents the actual delivery, measurable outputs, and community benefit achieved.

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	TRANSFORMATION
P0	Vision and Mandate	270	240	88.9%	226	GOOD	HIGH	Political commitment + consultation → 2050 vision, 4 missions, SDG mapping
P1	Governance and Coordination	250	210	84.0%	192	GOOD	MEDIUM	7 boards + Governance Board → coordinated cross-sector partner action
P2	Strategy and Program Design	265	240	90.6%	228	EXCELLENT	HIGH	Board mandates → City Leap IBP, Climate Strategy, LNRS, BCAI framework
P3	Capital Deployment	275	195	70.9%	164	FAIR	MEDIUM	£1.67bn budget → deployed capital in green infrastructure and services
P4	Infrastructure Delivery	260	220	84.6%	202	GOOD	HIGH	Deployed capital → heat networks, solar, retrofit, EV

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	TRANSFORMATION
P5 	Community and Household Uptake	240	140	58.3%	107	BOTTLENECK	LOW	charging, LED lighting
								Installed infrastructure + programs → household behaviour change, deprived-community benefit
P6	Outcome Measurement	270	248	91.9%	238	EXCELLENT	HIGH	City-wide data → verified progress against 2030 target, QoL indicators, SDGs

Sustainability-Linked Bond Opportunity

Bristol has issued two tranches of a Local Climate Bond (LCB) through Abundance Investment (Bristol Climate Action Investment, BCAI), raising approximately £2 million at 4.2% fixed over 5 years. This is a use-of-proceeds instrument: funds are ring-fenced for specific green projects (heat pumps, solar, LED lighting). While successful as a community engagement mechanism, the LCB does not provide access to institutional capital markets and has no mechanism to link Bristol's coupon cost to its performance against the four One City missions.

A Sustainability-Linked Bond (SLB), structured under the ICMA Sustainability-Linked Bond Principles (SLBP, June 2024), offers a fundamentally different proposition. The coupon adjusts (steps up) if Bristol misses pre-agreed Sustainability Performance Targets (SPTs) at defined observation dates. Proceeds are general purpose. Institutional investors (pension funds, insurance companies, ESG-mandated funds) can participate at scale: typical UK municipal SLB sizes range from £50m to £200m. Bristol's existing monitoring architecture (City Leap contractual KPIs, 256 QoL indicators, PHOF, BACCC annual reviews) provides the data infrastructure required for credible KPI selection and annual verification.

Proposed SLB KPI Framework (indicative)

KPI AXIS	PROPOSED KPI	BASELINE	2027 SPT	2030 SPT	STEP-UP IF MISSED	VERIFICATION SOURCE
Environment (E)	City-wide CO2 reduction vs 2005 baseline	43% (2023)	55%	70%	25 bps	City Council annual carbon report; BACCC review

KPI AXIS	PROPOSED KPI	BASELINE	2027 SPT	2030 SPT	STEP-UP IF MISSED	VERIFICATION SOURCE
Environment (E)	City Leap zero-carbon generation capacity (MW)	0 MW (2022 baseline)	120 MW	180 MW	25 bps	City Leap contractual KPI audit (Ameresco/BCC)
Social (S)	Male life expectancy gap (most vs least deprived)	10.7 years (2021-23)	Below 10.0 yr	Below 9.0 yr	20 bps	PHOF 2025 (A02a); ONS data
Social (S)	Residents in 10% most deprived areas	53,100 (11%, IMD 2025)	Below 10.5%	Below 9.5%	15 bps	IMD update; QoL Survey deprivation decile data
Governance (G)	City Leap social value delivered	£30m (of £61.5m, ~2024)	£50m	£61.5m+	10 bps	City Leap annual social value report
Note: KPI selection, SPT calibration, and step-up triggers require ICMA SLBP external review (Second Party Opinion) before issuance. The TrueValue N-D-C framework provides the independent phase-resolved scoring methodology for annual KPI verification.						

Phase Intervention Worksheet

Provisional disclaimer: the following recommendations are derived analytically from N-D-C phase data and publicly available sources. They represent a structural diagnostic, not a capital commitment recommendation. Field validation, stakeholder co-design, and independent due diligence are required before any capital or policy decision is taken.

Five-Axis Failure Diagnosis

All five axes are below the 80-point threshold, indicating systemic stress across all dimensions of the Bristol One City delivery chain. The most critical signals are from In2 (temporal: the 2030 deadline is 4 years away and household uptake is near-zero) and e (financial: no institutional-scale bond instrument is yet structured).

CONSTANT	WHAT IT MEASURES	CHAIN SCORE	SYSTEM DIAGNOSIS	INTERVENTION CATEGORY
π	Operational balance: chain average	81.3 (above 80)	Good aggregate, but Phase 5 failure is masked. Addressing Phase 5 alone would lift π to ~84.	Community engagement infrastructure; household programme design

CONSTANT	WHAT IT MEASURES	CHAIN SCORE	SYSTEM DIAGNOSIS	INTERVENTION CATEGORY
ϕ	Proportional value distribution	71.0 (below 80)	Phase 5 N=107 vs chain avg 194: value is not reaching communities proportionally.	Just transition fund; deprived-ward targeting; community wealth building
$\sqrt{2}$	Structural overhead: D>>C friction	65.0 (below 80)	D-C gap of 100 at Phase 5, 80 at Phase 3. Requirements are binding but delivery lags.	Programme simplification; trusted intermediaries; place-based neighbourhood model
ln2	Value transformation: temporal gap	61.0 (below 80)	6+ years since 2019 declaration; 2030 deadline 4 years away; household uptake near-zero.	Accelerated delivery programme; net-zero neighbourhood demonstrators; pay-as-you-save
e	Financial abstraction: instruments linked to real metrics	52.0 (below 80)	BCAI at £2m; no SLB structured; City Leap JV is not a bond instrument.	Sustainability-linked bond framework; ICMA SLBP alignment; institutional investor engagement

Bottleneck Phase: P5: Community and Household Uptake (Balance 58.3%)

Phase ID: P5 | **Balance score:** 58.3% (Poor zone, below 61.8% threshold) | **D-C gap:** 100 points (240 vs 140). This is the largest D-C gap in the chain.

D-C imbalance diagnosis: D=240 is driven by binding policy requirements (10% mandatory BNG, pay-as-you-save model designed, £1.5m community energy fund established, BCAI accessible from £5). C=140 reflects only 750 BCAI retail investors from a population of 470,000; scope 3 emissions structurally untracked; LNRS not covering most development sites; life expectancy gap not improving; community energy uptake in early stage. The binding requirements (D) are real but the programme design has not yet connected with the lived reality of deprived communities. This is a structural problem of trusted intermediary absence and programme accessibility, not a resource gap alone.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
Net-Zero Neighbourhood demonstrators	Co-design 3 demonstrator projects in Hartcliffe/Withywood, Filwood, and Lawrence Hill (Bristol's three most deprived wards) following the Living Places Earth Net-Zero Neighbourhood model: c.1,000 properties, blended finance stack, pay-as-you-save, no upfront cost to residents. Target: household uptake at scale via a trusted, locally-rooted intermediary rather than a retail investment platform.	C up by approximately 25-30 points (from 140 to 165-170): direct measurable uptake in 3,000 properties would represent ~0.6% of Bristol households, lifting C proportionally. D unchanged (existing requirements already in place). Estimated by analogy with comparable UK demonstrators (Innovate UK Mission Net Zero, Bristol's own programme). Source: Living Places Earth place-based one-pager; Bristol City Leap Mission Net Zero programme.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
Scope 3 household tracking programme	Introduce a structured city-wide consumption-based emissions tracking methodology at ward level, building on the BACCC 2023 recommendation to address the scope 3 data gap. Partner with the University of Bristol Cabot Institute to develop a QoL Survey supplement on consumption emissions (transport, food, goods). This lifts both C (new measurable output) and the Phase 6 data architecture.	C up by approximately 10-15 points (from 140 to 150-155): creating a measurable output where currently "structurally untracked" is the status raises C even before behaviour change occurs. Source: BACCC 2023 gap analysis (implied); QoL Survey methodology (256 indicators, extensible). This is an enabling intervention, not a direct delivery intervention.
Deprived-ward targeted BCAI tranche	Issue a third BCAI tranche specifically targeting residents of the 30 Bristol wards in the 10% most deprived nationally, with a below-market entry rate (e.g. £1 minimum, ISA-eligible, with council match-contribution for residents in receipt of Universal Credit). This directly addresses the geographic and financial exclusion evident in the current 750-investor BCAI base.	C up by approximately 8-12 points: at £500 average per deprived-ward investor and 10,000 participants (2% of 53,100 deprived residents), this raises BCAI to £5-7m and broadens the community ownership base measurably. D may increase modestly (~5 points) as new administration and equity monitoring requirements are added. Net D-C effect: gap narrows by approximately 15-17 points. Source: analogy with Camden Municipal Bond (2015) and Abundance's BCAI data.
Sustainability-Linked Bond (SLB) issuance	Structure and issue a £50-100m SLB under ICMA SLBP with 5 KPIs spanning the four One City missions (carbon, City Leap MW capacity, life expectancy gap, deprivation rate, social value delivered). Coupon step-up of 25-50 bps if SPTs missed at 2027 and 2030 observation dates. Proceeds general purpose (supplement PWLB borrowing). TrueValue N-D-C framework provides independent phase-resolved scoring for annual KPI verification. This is the e-axis intervention.	e-axis score up from 52 to approximately 72-75: an SLB at £100m would represent 5% of Bristol's capital budget, creating institutional accountability for mission delivery. The N-D-C scoring methodology becomes the verification backbone. D at Phase 3 increases modestly (~10 points, new reporting obligations). C at Phase 3 increases by approximately 25-35 points (actual institutional capital mobilised). Phase 3 balance would improve from 70.9% to approximately 78-80%. Source: ICMA SLBP June 2024; Enel SLB precedent; GFI LCB Toolkit.

Combined effect estimate: If all four interventions are applied, Phase 5 C would increase from 140 to approximately 185-195 (net of the neighbourhood demonstrators + scope 3 tracking + BCAI deprived-ward tranche). Balance would improve from 58.3% to approximately 71-75% (entering Fair zone, approaching Good). Phase 3 would improve from 70.9% to approximately 78-80% via the SLB capital mobilisation. Chain balance would lift from 81.3% to approximately 85-87%. The Phase 5 signal on the In2 axis (temporal gap) would not be resolved by these interventions alone: the 2030 deadline creates irreducible urgency that requires parallel acceleration across all four interventions simultaneously.

Axis cross-reference: Phase 5 bottleneck is primarily signalled by $\sqrt{2}$ (structural overhead: $D > C$ friction), $\ln 2$ (temporal gap: 6+ years to near-zero household uptake), and ϕ (proportional value: Phase 5 $N=107$ vs chain average 194). The SLB intervention addresses the e -axis directly and the $\sqrt{2}$ axis partially. Neighbourhood demonstrators address $\sqrt{2}$ and $\ln 2$.

Secondary Bottleneck: P3: Capital Deployment (Balance 70.9%)

Phase ID: P3 | Balance score: 70.9% (Fair zone, below 80%) | D-C gap: 80 points (275 vs 195).

D-C imbalance diagnosis: $D=275$ is the highest in the chain: £1.67bn approved capital budget, PWLB borrowing constraints, City Leap 50/50 JV governance, ICMA Green Finance Framework requirements. $C=195$ reflects ~£30m of £61.5m social value target delivered, BCAI at only £2m, and scope 3 capital mobilisation essentially zero. The governance and regulatory architecture is fully in place (D very high) but the capital has not yet reached the communities and assets that would raise C proportionally.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
SLB issuance (e-axis)	As described in Phase 5 worksheet above: £50-100m institutional SLB under ICMA SLBP. Proceeds supplement PWLB borrowing for green capital expenditure aligned to City Leap and One City missions.	C up by 25-35 points (from 195 to 220-230): institutional capital mobilised at scale. Phase 3 balance would improve from 70.9% to approximately 78-82%. Source: ICMA SLBP; GFI LCB Toolkit; Room151 PWLB alternatives analysis.
City Funds impact capital deployment	Engage City Funds (represented on the Governance Board by Andy Street) to deploy matched impact investment alongside the SLB into the Net-Zero Neighbourhood demonstrators. Blended finance stack: SLB (senior) + City Funds (mezzanine) + community pay-as-you-save (junior).	C up by 10-15 points: adds a local anchor institution's capital to the blended stack, improving both scale and community accountability. D increases modestly (blended finance administration). Net effect: P3 gap narrows by approximately 25-30 points. Source: Living Places Earth blended finance model; City Funds governance remit.

Axis cross-reference: Phase 3 bottleneck is primarily signalled by e (financial abstraction: no SLB, BCAI only £2m) and $\sqrt{2}$ (structural overhead: high governance D with slow capital C).

Risk Factors and Opacity Sources

RISK FACTOR	PHASE	D OR C	STRUCTURAL CAUSE (NOT ATTRIBUTION)
Scope 3 emissions untracked	P5, P6	C (missing output)	Consumption-based emissions (transport, food, goods) require household-level data collection methodology not yet implemented. National statistical infrastructure does not provide city-level scope 3 data.

RISK FACTOR	PHASE	D OR C	STRUCTURAL CAUSE (NOT ATTRIBUTION)
2030 carbon target out of reach	P0, P5	D-C gap widening	BACCC 2023 progress review noted the 2030 net-zero (scope 1/2/3) target is structurally very challenging given the pace of household and scope 3 behaviour change. The target itself (D) is fixed; C is not accelerating fast enough to close the gap.
LNRS focus area coverage gap	P4, P5	C (limited reach)	West of England LNRS focus areas are parks, riverbanks, and railway margins, none of which are likely development sites. Mandatory BNG strategic significance uplift therefore does not apply to most Bristol development. Structural, not a policy failure: the LNRS mapping methodology excludes pre-existing habitats.
City Leap scope 3 social value gap	P3, P4	C (partial delivery)	£30m of £61.5m social value target delivered at approximately the midpoint of the 5-year term. The social value model relies partly on contracts delivered by local supply chain (£50m target) which requires a mature local SME pipeline not yet fully developed.
Inter-board integration nascent	P1	C (coordination gap)	The four One City missions were only adopted in October 2024. Cross-board integration is structurally new: the 7 boards have historically operated within their own thematic lanes. Mission-based integration requires new working patterns that take time to embed.

Data Sources

Bristol One City Plan 2025 (5th iteration): bristolonecity.com: Four missions, governance structure, 200+ stakeholders

Bristol One City Climate Strategy (Arup, February 2020): 10 delivery themes, 2030 net-zero scope 1/2/3: [PDF](#)

Bristol City Leap Initial Business Plan Summary (April 2024): £631m, contractual KPIs: [PDF](#)

Bristol Climate Action Investment (Abundance Investment, March 2025): 4.2% LCB, Green Finance Framework: abundanceinvestment.com

Bristol City Council Performance Framework 2025-2028: Tier 1/2 metrics, Power BI dashboard: bristol.gov.uk

Bristol Key Facts November 2025: population, deprivation, health: bristol.gov.uk

Quality of Life Survey 2024-25: 256 indicators, 34 wards, 10 deprivation deciles: bristol.gov.uk

Bristol Voluntary Local Review 2022: 140+ SDG indicators: [UN DESA](https://un.org/sdgs)

West of England Local Nature Recovery Strategy (WENP, November 2024): LNRS focus areas: wenp.org.uk

ICMA Sustainability-Linked Bond Principles (June 2024): 5 core components: icmagroup.org

Bristol Advisory Committee on Climate Change (BACCC): 12 recommendations (2023 progress review): bristolclimatenature.org

Living Places Earth: place-based net-zero advisory, David White (former City Leap Operations Director): livingplaces.earth

- ← **Hub Overview**
- Supply Chain Overview**
- N-D-C Dashboard**
- Return Chain**
- TrueValue Home**

TVPCI-R: JUST TRANSITION RETURN CHAIN

Bristol One City: Community Wealth Return Analysis

The TVPCI-R measures how much of the One City investment and benefit recirculates to Bristol's most deprived communities. TVPCI-R = 50.7 (Failure zone). B_chain = 47.0 (severe imbalance). The gap between the primary delivery chain (TVPCI 81.3%) and the community return chain (50.7%) is the defining structural failure of the Bristol system. All values PROVISIONAL.

81.3

TVPCI (primary chain balance)

50.7

TVPCI-R (community return score)

47.0

B_chain (holon balance)

+30.6

Delta_chain (value flowing out, not circulating back)

Framework: TVPCI, TVPCI-R, and B_chain

In the Tholonic framework, the primary delivery chain (TVPCI) measures D-C balance: how well the institutional architecture (D) is matched by actual delivery (C). The return chain (TVPCI-R) measures a different question: of the value generated by the delivery chain, how much circulates back to the communities that most need it? For Bristol, this is the "just transition" dimension: the degree to which the decarbonisation transition also reduces inequality and improves outcomes for the 53,100 residents in the most deprived 10% nationally.

D ROLE: DEFINITION**TVPCI (Primary Chain)**

How well does the institutional delivery chain balance its requirements (D) against its outputs (C)? Measures the coherence of the city's delivery system as a whole.

81.3 (Good)**C ROLE: CONTRIBUTION****TVPCI-R (Return Chain)**

How much of the investment and benefit generated by the delivery chain recirculates to deprived communities? Measures the just transition dimension: the "ecological return" of social equity.

50.7 (Failure)**N ROLE: NEGOTIATION****B_chain (Holon Balance)**

The emergent coherence of the whole system: how well the primary chain and the return chain are in balance. A high B_chain means the city is both delivering and returning value proportionally. $B_chain = 100 \times \exp(-2 \times |TVPCI - TVPCI-R| / \max(TVPCI, TVPCI-R))$.

47.0 (Severe imbalance)

Delta_chain = TVPCI - TVPCI-R = +30.6 (positive: value is flowing forward through the institutional chain but not circulating back to deprived communities)

TVPCI (primary chain)

81.3%

TVPCI-R (return chain)

50.7%

Per-Phase Return Scores (R_p) and Omega Weights

All R_p scores and omega weights are PROVISIONAL. R_p measures how observable and accountable each phase's community return is. Omega weights are proportional to community impact potential at each phase and must sum to 1.0. Zone thresholds: Coherent R_p ≥ 80; Stressed 61.8-79; Failure 38.2-61.8; Breakdown < 38.2.

P0: VISION

R_p = 62

STRESSED

ω = 0.08

Plan names equity explicitly. QoL disaggregated by deprivation. But no enforcement of equity requirements in mandate text.

P1: GOVERNANCE

R_p = 55

FAILURE

ω = 0.08

VOSCUR (voluntary sector) and TUC represented on Governance Board. Community voice is present but outweighed by institutional anchors. Co-design of missions limited.

P2: STRATEGY

R_p = 60

FAILURE

ω = 0.10

City Leap has £61.5m social value target incl. local supply chain. BCAI accessible from £5. But strategies are technically-authored, not co-designed with deprived communities.

P3: CAPITAL

R_p = 48

FAILURE

ω = 0.18

~£30m of £61.5m social value delivered. Lawrence Weston community energy. BCAI at £2m. Most capital flows to institutional partners, not deprived communities directly.

P4: INFRASTRUCTURE

R_p = 65

STRESSED

ω = 0.22

Solar at schools, housing retrofit focused on social housing, LED lighting in public buildings. Direct benefit to deprived communities

P5: COMMUNITY UPTAKE ⚠

R_p = 35

BREAKDOWN

ω = 0.25

750 BCAI investors from 470k population. Hartcliffe, Filwood, Lawrence Hill not visibly benefiting. Scope 3 unaddressed.

P6: OUTCOMES

R_p = 40

FAILURE

ω = 0.09

QoL survey disaggregated by deprivation. But life expectancy gap 10.7yr not improving. Mental wellbeing gap persistent. Outcomes

is real but geographically narrow.

The primary community interface is essentially absent.

do not yet reflect community wealth return.

TVPCI-R Calculation

PHASE	R_P	Ω (OMEGA)	Ω × R_P	ZONE
P0: Vision and Mandate	62	0.08	4.96	STRESSED
P1: Governance	55	0.08	4.40	FAILURE
P2: Strategy Design	60	0.10	6.00	FAILURE
P3: Capital Deployment	48	0.18	8.64	FAILURE
P4: Infrastructure Delivery	65	0.22	14.30	STRESSED
P5: Community Uptake Δ	35	0.25	8.75	BREAKDOWN
P6: Outcome Measurement	40	0.09	3.60	FAILURE
TOTAL		1.00	50.65	FAILURE

TVPCI-R = Σ(ω × R_p) = 50.7 | B_chain = 100 × exp(-2 × |81.3 - 50.7| / 81.3) = 100 × exp(-0.754) = 47.0

Community Impact Assessment: Who Receives the Return?

The omega weights above reflect community impact potential: Phase 5 carries the highest weight (0.25) because it is the direct community interface, and Phase 4 (0.22) because infrastructure physically enters deprived communities. The collapse of R_p at Phase 5 (35, Breakdown) combined with its high omega weight is the primary driver of the low TVPCI-R.

COMMUNITY DIMENSION	CURRENT STATUS	RETURN CHAIN SIGNAL	PHASE(S)
Financial participation	750 BCAI investors from 470,000 population (0.16%)	Breakdown: financial return is not reaching deprived communities	P3, P5
Health inequality	10.7yr male life expectancy gap (most vs least deprived, 2021-23)	Failure: health returns not reflected in outcome data despite significant investment	P5, P6

COMMUNITY DIMENSION	CURRENT STATUS	RETURN CHAIN SIGNAL	PHASE(S)
Mental wellbeing	24.4% poor mental wellbeing (most deprived) vs 11.5% (least deprived)	Failure: wellbeing gap persistent, not narrowing	P5, P6
Physical infrastructure access	Solar at schools, housing retrofit in social housing, LED lighting in public buildings	Stressed: tangible but geographically narrow; 30 deprived wards not fully covered	P4
Energy cost benefit	Pay-as-you-save model designed but not yet deployed at scale	Failure: the financial benefit of the energy transition has not yet reached households in fuel poverty	P3, P5
Employment (local supply chain)	£30m of £61.5m social value target delivered; £50m local supply chain target	Failure: approximately 49% of 5yr social value target delivered at midpoint; local SME pipeline still developing	P3, P4
Scope 3 / consumption equity	No structured city-wide scope 3 reduction programme at household level	Breakdown: the consumption emissions of deprived communities (often lower per capita) are not being tracked or supported	P5, P6

Priority Just Transition Interventions (Bond-Linked)

The following interventions are the highest-priority actions to close the TVPCI-TVPCI-R gap. Each is structured to be measurable, phase-specific, and bondable (i.e. capable of serving as a KPI or SPT in a Sustainability-Linked Bond or Local Climate Bond tranche).

PHASE	INTERVENTION	DESCRIPTION	EXPECTED R_P IMPACT	BOND/FINANCE LINK
P5 (primary)	Net-Zero Neighbourhood demonstrators in 3 deprived wards	Co-design and fund net-zero retrofit of c.1,000 properties each in Hartcliffe/Withywood, Filwood, and Lawrence Hill (Bristol's three most deprived wards). Blended finance: SLB (senior) + City Funds (mezzanine) + pay-as-	R_p P5: 35 → approximately 55-60 (+20-25 pts). Omega = 0.25. TVPCI-R impact: +5-6 pts overall.	SLB SPT: number of properties retrofitted in IMD top-10% wards. Observation date: 2027 (1,000 props), 2030 (3,000 props).

PHASE	INTERVENTION	DESCRIPTION	EXPECTED R_P IMPACT	BOND/FINANCE LINK
		you-save (junior). Living Places Earth model.		
P3 (capital)	Deprived-ward BCAI tranche with council co-contribution	Third BCAI tranche at £1 minimum entry, ISA-eligible, with a council match-contribution for Universal Credit recipients. Target: 10,000 investors from the 53,100 in the most deprived decile. Raise BCAI total to £5-7m.	R_p P3: 48 → approximately 58-62 (+10-14 pts). Omega = 0.18. TVPCI-R impact: +2-3 pts.	SLB SPT: number of BCAI investors from IMD top-10% postcode areas. Verification: Abundance Investment investor postcode data.
P5 and P6	Scope 3 ward-level consumption tracking	Partner with Cabot Institute to add a consumption-based emissions supplement to the annual QoL Survey. Create ward-level scope 3 tracking for 34 wards, disaggregated by deprivation decile. Establishes the data layer needed for scope 3 SPTs in a future SLB.	R_p P5: +8-10 pts (new measurable output). R_p P6: +10-12 pts (data quality improvement). Combined TVPCI-R impact: +3-4 pts.	Pre-condition for scope 3 SLB KPI. Bond-linkable once methodology established and first data published.
P3 and P5	SLB issuance with life expectancy gap SPT	Issue a £50-100m Sustainability-Linked Bond with the male life expectancy gap (most vs least deprived wards) as a Social SPT alongside the environmental KPIs. 25 bps step-up if gap exceeds 10.0 years at 2030 observation. TrueValue N-D-C scoring provides annual independent verification of all SPTs.	R_p P3: +15 pts (institutional capital with equity accountability). TVPCI-R systemic impact: +4-6 pts when all KPIs are linked to community return. B_chain: 47.0 → approximately 58-62.	Direct SLB SPT: male life expectancy gap at 2027 and 2030 observation dates. Coupon step-up of 25 bps if missed. Verification: PHOF (A02a), ONS data.

Calculation Reference

R_p = community return score for phase p (0-100): how observable and accountable the phase's community return is. Ω_p = phase weight (proportional to community impact potential; sums to 1.0). $TVPCI_R = \sum(\Omega_p \times R_p)$. $B_{chain} = 100 \times \exp(-2 \times |TVPCI - TVPCI_R| / \max(TVPCI, TVPCI_R))$. Zone thresholds (phi-derived): Coherent $R_p \geq 80$ | Stressed 61.8-79 | Failure 38.2-61.8 | Breakdown < 38.2.

METRIC	VALUE	ZONE	INTERPRETATION
TVPCI (primary chain)	81.3	GOOD	The institutional delivery chain is functioning well in aggregate, with Phase 5 as the exception.
TVPCI-R (return chain)	50.7	FAILURE	Community wealth return is in the Failure zone. No phase achieves Coherent status.
B_chain (holon balance)	47.0	BREAKDOWN	Severe imbalance between primary delivery and community return. The system is not in sustainable equilibrium.
Delta_chain (TVPCI - TVPCI-R)	+30.6	Imbalanced	Positive delta: value is flowing forward through the institutional chain but not recirculating back. This is the structural definition of an unjust transition.
Weakest return phase	P5: $R_p = 35$ (Breakdown)	BREAKDOWN	Community and household uptake is the primary bond intervention candidate for the return chain.

VALUE CHAIN

Bristol One City: Sustainability-Linked Bond Value Layer

Value chain financial data not yet available. The SLB KPI framework, ICMA SLBP alignment, SPT calibration, and coupon step-up design require Phase 5 intervention data and institutional investor engagement before this layer can be activated.



Value Chain Analysis: Deferred

The Bristol One City value chain centres on the structure of a Sustainability-Linked Bond (SLB) that would translate the One City delivery chain into an institutional-capital instrument. This layer is deferred pending completion of three prerequisites listed below.

When activated, this layer will include: SLB KPI selection (5 axes: E, S, G, mapped to One City missions); Sustainability Performance Target (SPT) calibration against City Leap contractual KPIs and QoL Survey baselines; bond characteristic design (size, tenor, coupon step-up triggers at 2027 and 2030 observation dates); ICMA SLBP Second Party Opinion requirements; and an annual TrueValue N-D-C verification framework mapped to the 7-phase delivery chain.

The e-axis coherence score of 52.0 (the lowest in the five-model framework) confirms that this layer is the critical gap in the Bristol system. The supply chain and return chain analyses provide the quantitative foundation; this layer converts that foundation into an investable financial product.

Prerequisites for Value Chain Activation

- Phase 5 neighbourhood demonstrator data: at least one Net-Zero Neighbourhood pilot operational with measurable household uptake data (R_p and omega calibration)
- Scope 3 tracking methodology established: ward-level consumption emissions data published (first QoL Survey supplement complete)

→ Institutional investor engagement: at least one pension fund or insurance company investor expressing interest in Bristol SLB at indicative terms, confirming market appetite for the SPT structure

→ ICMA SLBP external review (Second Party Opinion) commissioned: KPI materiality and SPT ambition level independently confirmed

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